

Algorithm: K-Nearest Neighbors (KNN)**Dataset: Breast Cancer Dataset (from sklearn.datasets)**

- *“Created a new notebook with custom implementation”*

<u>Task</u>	<u>Actions Taken</u>	<u>Rationale</u>	<u>Difficulty Level (1-10)</u>	<u>Time Taken</u>
Data Understanding	Explored Breast Cancer dataset, checked features, and identified target labels	To understand data structure and ensure proper preprocessing	3	20 min
Data Preprocessing	Split dataset (80/20), applied StandardScaler for normalization	KNN is distance-based, so feature scaling is necessary	4	30 min
Implemented Custom KNN	Coded KNN algorithm using Euclidean distance, loops, and sorting	To understand the inner workings of KNN and compare with Sklearn's implementation	8	1 hour 30 min
Implemented Sklearn KNN	Used KNeighborsClassifier from Scikit-learn	To compare performance with custom implementation	3	20 min
Hyperparameter Tuning (k-values)	Tested k values (3, 5, 10, 15) and recorded accuracy	To find the optimal k value for the best classification performance	5	50 min
Model Evaluation	Compared accuracy of custom and Sklearn KNN models	To assess effectiveness and performance trade-offs	4	30 min

Conclusion & Analysis	Observed underfitting at k=15, best accuracy at k=10	To summarise findings and determine practical implications	3	20 min
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1. Business Understanding

Objective:

The goal of this work is to implement and evaluate the K-Nearest Neighbors (KNN) algorithm, both from custom and using the built-in Scikit-learn library. We will be comparing the performance of these two approaches on the Breast Cancer Dataset to determine the feasibility of using a custom implementation in real-world applications. The KNN algorithm is a simple, instance-based learning algorithm that classifies new samples based on the majority class of their closest neighbors in the feature space. KNN is intuitive and easy to implement, but its performance is highly dependent on the choice of k, the number of neighbors to consider. The comparison will highlight the accuracy and efficiency differences between the custom and library KNN models. The analysis will also investigate the sensitivity of the KNN algorithm to the value of k.

2. Data Understanding

Dataset Used: Breast Cancer dataset from **sklearn.datasets** (includes features like mean radius, mean texture, mean smoothness, etc.)

Feature Description:

- The dataset contains **30 numerical features** that represent various measurements of the breast cancer cell nuclei.
- Target: The target variable is a binary class (0 for malignant and 1 for benign tumors).
- Initial Observations:
 - The dataset has 569 samples with 30 features, and the target variable has two distinct classes (benign and malignant).
 - The data is relatively clean, but standardization of features is often necessary for distance-based algorithms like KNN.

3. Data Splitting

Modifications Made:

The dataset was split into training and test sets using an 80/20 split.

Feature scaling (standardization) was performed using **StandardScaler** from sklearn to normalize the features. Standardization ensures that all features have a **mean of 0** and a **standard deviation of 1**, making them comparable when calculating distances.

Random state 42 was used to ensure **consistent data splits** for reproducibility.

Potential Issues:

None were identified in this stage, but further tuning of k and possible handling of imbalanced classes may improve results.

4. Modelling (KNN from custom and Sklearn)

Implementation Details:

The custom KNN model was implemented from custom using Euclidean distance as the metric for determining the nearest neighbors.

k=3 neighbors were selected initially for the custom KNN model.

The model was trained using the training data and predictions were made using the test data.

5. Evaluation

The models were evaluated using the accuracy score metric, and a comparison was made based on the following results:

<i>K Value</i>	Custom KNN Accuracy	Library KNN Accuracy
3	0.93860	0.9275
5	0.93860	0.9319
10	0.95614	0.9363
15	0.94737	0.9231

6. Conclusion

Custom KNN implementation able slightly outperform Sklearn implementation. Custom & Library Models Agree: K=10 provides the highest accuracy. When K=15 Shows Slight Underfitting, Accuracy drops slightly after K=10.